

Cross-Lingual Transferability of Training Data Extraction Attacks to Recover Memorized PII

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Abstract—The robustness of Personally Identifiable Information (PII) protection in Large Language Models (LLMs) is a critical concern, yet the risks associated with cross-lingual data extraction remain under-explored. This study evaluates the vulnerability of English-centric and multilingual models to Training Data Extraction (TDE) attacks when prompted in non-English languages.

We construct a multi-domain PII dataset comprising social media handles, email addresses, and phone numbers and translate the attack contexts into Italian, Spanish, French, and German. Our results show that TDE attacks against both English-centric and multilingual models transfer to different languages: the attacks are successful on translated prompts, even though only the original English prompt might have been included in the pre-training data. A web-presence check on a sample of the translations confirms that they are not available online. The share of English leaks recovered in other languages grows with the multilingual capability of the model, and it drops sharply when the original wording is lost, even without a change of language.

This suggests that native multilingual pre-training facilitates the emergence of latent cross-linguistic bridges that simplify the retrieval of personally identifiable information (PII). We analyze the activations of multilingual large language models (LLMs) and find that different translations of the same prompt are bridged in similar representations, with the strongest alignment in the middle layers.

Our results highlight a fundamental security gap in modern LLMs, necessitating more robust, language-agnostic sanitization strategies for future model alignment.

Index Terms—training data extraction, transferability, cross-lingual evaluation

I. INTRODUCTION

Large Language Models (LLMs) are known to memorize fragments of their training data, and this memorization can be exploited to extract personally identifiable information (PII) – like emails, phone numbers, addresses, credentials – through Training Data Extraction (TDE) attacks [1], [2]. Most work on this threat, however, treats memorization as a monolingual phenomenon: an attacker is assumed to query a model in the same language the memorized text was originally written in.

This assumption is broken down for the multilingual LLMs now deployed at scale, leaving a consequential blind spot. If a model, trained on multilingual data, **memorizes some data in one language** and can be prompted to **regurgitate that same memorized content when queried in other languages**, then privacy safeguards evaluated only in the source language systematically underestimate real-world exposure. Moreover,

as multilingual capability improves, it may not only add new attack surfaces but also actively strengthen extraction across all of them if the model is building shared cross-lingual representations that link the same fact regardless of the language in which it is expressed.

Despite these stakes, whether TDE attacks transfer across languages and, if so, what mechanism underlies that transfer remains largely unexplored. Prior work has separately established that LLMs memorize and leak training data [1]–[4], and safety alignment can be circumvented via cross-lingual prompting [5], but these two lines of research have not been connected. It is not known whether memorization itself is language-agnostic (i.e., whether a fact memorized from an English document is stored in a way that a differently-worded, differently-langauged prompt can still trigger), or whether apparent leakage in a second language is simply an artifact of translation overlap with other multilingual training data. Answering this is what motivates our approach.

In this paper, we study **memorization in a cross-lingual scenario**: given an English dataset highly likely to appear in the training data of both monolingual and multilingual models, we ask whether data memorized from an English document can be extracted using an attack formulated in another language. We use `Pile-CC` a component of `Pile` [6] derived from Common Crawl. Because `Pile`’s collection was fully automated, it inadvertently contains substantial PII embedded in scraped blogs, forums, and news pages. We translate paragraphs from `Pile-CC` into multiple languages and apply TDE attacks in each target language, testing whether English training produces leakage when probed cross-lingually.

Our experiments show that both **monolingual and multilingual models are susceptible to this cross-lingual leakage**, and that the risk increases with the model’s multilingual capability. These results point to the existence of **cross-lingual bridges**: shared underlying representations that allow attackers to recover memorized secrets even when the attack language differs from the language of the original training data. We further show, by comparing strict translations against paraphrases in both languages, that this leakage tracks fidelity to the memorized content and localize the effect to a specific band of middle transformer layers.

II. BACKGROUND AND RELATED WORK

This section reviews training data extraction as a privacy threat, the defenses proposed against it, and the still-limited evidence on whether either the threat or the defense transfers across languages.

a) Privacy Vulnerabilities and Training Data Extraction: Unintended memorization of the training data allows an attacker to extract such training data. [1] first showed how an attacker could exploit such phenomena: querying GPT-2 with training-set prefixes recovers verbatim PII, a phenomenon they formalized as k -eidetic memorization. [2] later showed memorization scales log-linearly with model capacity and duplication frequency, and that a matching training-context prefix substantially increases extraction success – a finding echoed by [3] for emails and social handles. [4] extended this beyond controlled settings, showing open models like GPT-Neo and Pythia leak more than earlier estimates suggested, while [7] confirmed that closed, production-grade models such as GPT-4 remain vulnerable to black-box extraction.

These studies, however, focus primarily on monolingual contexts centred on English. Nevertheless, cross-linguistic vulnerabilities have already emerged in terms of robustness against adversarial attacks: [5] demonstrated that security alignment can be circumvented by translating jailbreak prompts into languages with scarce resources, implying that LLMs share a cross-linguistic latent space in which a security measure applied in one language is not necessarily valid in another. It has not been directly verified whether the same holds true for *extraction* as opposed to jailbreaking.

b) Strategies to Protect Privacy in LLMs: Mitigations can be applied at the data level, at training time, or as post-hoc model transformations. Data-level approaches, from classical NER [8] to transformer-based rewriting [9], scrub PII before training, but have been shown to scale poorly to massive, uncurated corpora [10]. Training-time approaches such as Differential Privacy [11] offer formal guarantees but are costly and degrade the utility of the trained model. Post-hoc approaches, like machine unlearning [12] and model editing [13]–[15], remove memorized content after the training has finished. Critically, all of these defenses are mostly evaluated monolingually: whether unlearning or editing a PII record in English also prevents its recovery when the model is queried in another language remains largely untested.

c) Cross-Lingual Memorization and Transferability: A small but growing body of work suggests the answer is no – privacy behavior does not transfer cleanly across languages, though the direction and mechanism of this failure are still unclear. [16] analyze memorization across 95 languages and show that the standard assumption – memorization tracks training-data volume – breaks down once cross-lingual similarity is modeled: among typologically similar languages, those with *less* training data are disproportionately more memorized, implying memorized content propagates through shared multilingual representations. [17] provide a direct behavioral demonstration of this: probing LLMs with book samples

machine-translated into languages that have no prior published translation, they find substantial recall (GPT-4o identifies title and author 69% of the time on entirely novel translations), showing that content memorized in English remains accessible under queries formulated in a language the model has never seen that content in.

At the same time, this transfer is demonstrably uneven. [18] show that unlearning applied in one language does not propagate to others – a fact deleted in English typically survives, verbatim, in Spanish or Chinese – indicating that whatever shared representation enables cross-lingual leakage does not straightforwardly enable cross-lingual forgetting. [19] and [20] further show that susceptibility to extraction and membership-inference attacks varies systematically with a language’s structural properties (redundancy, morphological complexity, resource level). Together, these results position language not as a neutral channel but as an active variable shaping both leakage and its mitigation.

Our contribution. Existing work thus establishes that memorized content can, in principle, cross language boundaries [16], [17], but this transfer is asymmetric and language-dependent [18]–[20]. What remains untested is the specific, practically consequential question this paper addresses: whether PII memorized from English training data text [17] – can be recovered via TDE attacks formulated in a different language. By translating paragraphs of Pile-CC and querying models in Italian, Spanish, French, and German for PII memorized only in English, we test directly whether the linguistic barrier offers any practical protection, and provide evidence for the **cross-lingual bridges** that may facilitate the recovery of secrets across linguistic boundaries. We demonstrate that such bridges are facilitated by high content fidelity and we localize their formation to a specific band of middle transformer layers.

III. METHODOLOGY AND EXPERIMENTS

The methodology of this study follows a multi-stage pipeline designed to evaluate the susceptibility of LLMs to cross-lingual Training Data Extraction (TDE).

A. Attack Methodology

We implement Training Data Extraction (TDE) attacks [1], [2], [4]. The objective is to determine if the models can verbatim generate a PII when prompted with a specific context.

1) Formalization of the TDE Attack: The objective of a TDE attack is to recover sensitive sequences $s \in \mathcal{D}_{train}$ [1]. Let θ represent the parameters of a causal language model M . The model defines a probability distribution over a sequence of tokens as

$$P_{\theta}(x_1, \dots, x_n) = \prod_{i=1}^n P_{\theta}(x_i | x_{<i}). \quad (1)$$

We select English samples from Pile-CC, and partition the sequence into a prefix p of length k and the target PII $\hat{s}$. The attack is successful if:

$$\hat{s} = \arg \max_{s'} P_{\theta}(s' | p) \quad (2)$$

that is, we define a sequence as *verbatim memorized* if the generated output $\hat{s}$ matches the ground truth s with an edit distance of zero. In this study, we fix the context window to $k = 200$ tokens. This specific length was chosen to provide a strong signal, as shorter contexts are also demonstrated to be less likely to extract memorized information [3], [14], [15].

The worst-case scenario for this attack is that the prompt p is derived directly from the target model’s training data: the success rate of such attacks is defined as *extractable memorization* [2].

2) *Cross-Lingual Mapping Challenge*: We compare the accuracy of semantically identical attacks, translated in different languages, to understand whether **memorized information in English can also be extracted with attacks in other languages**. This setup assesses whether the model’s internal semantic mapping allows for the leakage of sensitive data across language boundaries. While the original definition of *extractable memory* assumes that leakage is more likely when using exactly the same data that the model has seen – and potentially memorized – during the pre-training phase, we demonstrate that the attack is transferable to other languages.

B. Cross-Lingual Dataset Construction

Our dataset is composed of text samples from Pile-CC¹ that contains a PII. We collected three different PII types: email addresses, phone numbers and twitter handles. The scraped dataset, completely in English dataset, is composed of a *prompt* is followed by a *PII*.

To measure transferability of the attacks to other languages, we then machine translated each of the prompts in French, German, Italian and Spanish. The translation was performed using a Gemma 3 27b it via the Google Gemini API, instructing the model to respect the exact meaning of the original English sample and keeping the same formatting, and overall structure of the original prompt, to minimize the difference between the original English prompt and its translation. At least one author is proficient in each of the target languages, and spot checks were conducted to ensure the quality of the translations. At the end of this process, for each of the target languages, we obtained a dataset containing 4695 email, 9005 twitter handles and 3000 phone numbers.

1) *Novelty of the Translations*: To test whether the TDE attacks actually **transfer**, we need to verify that the **translations of the Pile-CC data are unlikely** to be used as part of the **pre-training** material of our tested model. While this seems reasonable, as we use specific samples from Pile-CC, we need to verify that such data are not already seen by the target models to ensure that we are measuring the **transferability** of the TDE attacks.

For most of the models tested, however, it is not possible to accurately verify the data used for training, since the model providers do not disclose the data used for pre-training. We therefore **search for instances of our translated data in search engines** and **check for matches** between our

translation and segments of the returned pages. If no matches are found, this means that the observed translation is unlikely to be available online and, consequently, unlikely to have been used for the pre-training phase of the target models.

To assess whether the translated versions of the English samples were publicly available online before their generation, we develop a web-based verification pipeline. We apply it to a subset of 200 translated samples per PII type. For each of these samples, we first generate a distinctive exact-match search query and use the Tavily search API² to identify potentially matching web pages. The URLs returned by the search engine are then compared with the Common Crawl index to identify archived versions that existed before the translation. For each relevant capture, we retrieve only the corresponding WARC record using HTTP range requests, extract the textual content of the page, and compare it with the translated sample. To compare the retrieved pages with the translations of our dataset, we compute exact matches and fuzzy matches.

Exact matching checks whether the original translated sample occurred verbatim in the extracted webpage text. *Fuzzy matching* computes the normalized Levenshtein-based similarity between the translated sample and candidate text windows of comparable length ratio from the webpage; a match was considered positive when the similarity score exceeded a predefined threshold (90%). This procedure provides evidence about whether the translated samples were publicly available on the web before their generation.

C. Target Models: English-centric and Multi-lingual models

We evaluate the attack across three distinct families of models to observe how scale and pre-training objectives influence cross-lingual leakage:

- EleutherAI GPT-Neo 1.3B and 2.7B [21] and GPT-J 6B [22]: English-centric models, these serve as a baseline to test the persistence of memorization when the model is forced to switch from its dominant training language (English) to a non-native prompt context. We selected these models because they are known to have been trained on The Pile, and therefore on the Pile-CC subset from which our evaluation samples were drawn.
- Meta Llama 3.2 1B and 3B [23]: multilingual models, these are integrated to evaluate the attack surface in models natively trained on a massive cross-lingual corpus, potentially facilitating the bridge between translated prompts and English-stored PII.
- Qwen Qwen 2.5 3B and 7B [24]: multilingual models trained on a large-scale, highly multilingual corpus. These models allow us to assess whether the observed cross-lingual memorization behavior generalizes beyond the Llama family and to compare the impact of different pre-training data mixtures and tokenization strategies on privacy leakage.

For Llama and Qwen models, there is no public confirmation that our selected Pile-CC subset was included in the

¹Original data available here, and verified by us scraping Pile-CC

²Documentation available on this website.

training data; nevertheless, the empirical evidence obtained from our attacks suggests that at least part of the evaluated content was likely seen during pre-training.

D. Finding Cross-Lingual Bridges

Let $x_n^{l,\text{eng}}$ and $x_n^{l,\text{target}}$ denote the hidden-state representations at layer l and token position n , where n corresponds to the last token of the input sequence. If a cross-lingual TDE attack succeeds, we hypothesize that models construct a cross-lingual bridge between representations in English (eng) and the corresponding representation in another language (target).

If cross-lingual TDE succeeds because the model has **learned a shared, language-agnostic representation of a specific memorized fact**, such as the association between a person’s name and their email address, then this representation should be recoverable whenever the **content is preserved**, regardless of the surface language. Conversely, if apparent cross-lingual leakage is driven by weaker signals, such as general topical or semantic similarity between the prompt and the training context, then any prompt conveying a similar meaning should activate comparable internal states, regardless of whether it precisely preserves the original content.

To distinguish these two accounts, we vary two factors independently: **language** and **wording fidelity**. We hence compare our strict, structure-preserving translations with free paraphrases. Crossing these factors allows us to determine not only whether translation moves a representation away from the original, but also whether this movement is driven by the language change itself or by the accompanying change in wording. Symmetrically, we can assess whether paraphrasing degrades the representation even when the language remains unchanged.

1) *Experimental Conditions*: For each sample x^{eng} in our dataset, we construct three additional variants:

- $x_{\text{strict}}^{\text{target}}$: a faithful, structure-preserving translation used in our main TDE attacks (see III-B). It introduces the same language change as the attack, with no wording changes beyond those required by translation.
- $x_{\text{para}}^{\text{eng}}$: an English paraphrase of x^{eng} . It preserves the language while changing the wording, thereby isolating the effect of rewording without a language change.
- $x_{\text{para}}^{\text{target}}$: a paraphrase of x^{eng} in the target language target. This condition changes both language and wording, and most closely resembles the setting of an attacker who does not have access to the exact original phrasing.

All paraphrases preserve the topic and general meaning of the source sample while deliberately varying lexical choice, sentence structure, and the order in which entities are introduced. Details on the generation and evaluation of the paraphrases are detailed in Appendix A.

We feed each variant through the target model and extract hidden states at every layer l , at the last token position n of the input prompt, immediately preceding the point where the PII would be generated. This representation captures the model’s accumulated contextual state, and has been demonstrate to effectively encode the contextual information of the input [25].

2) *Similarity metric*: We experimented with the cosine similarity as the similarity metric for the representations:

$$\frac{x_n^{l,a} \cdot x_n^{l,b}}{\|x_n^{l,a}\|, \|x_n^{l,b}\|}. \quad (3)$$

For each sample, we compute three per-layer similarity scores with respect to the English original context:

$$\text{sim}_{\text{strict}}^l = \text{sim}^l(x_n^{l,\text{eng}}, x_{n,\text{strict}}^{l,\text{target}}) \quad (4)$$

$$\text{sim}_{\text{para-eng}}^l = \text{sim}^l(x_n^{l,\text{eng}}, x_{n,\text{para}}^{l,\text{eng}}) \quad (5)$$

$$\text{sim}_{\text{para-target}}^l = \text{sim}^l(x_n^{l,\text{eng}}, x_{n,\text{para}}^{l,\text{target}}). \quad (6)$$

We then average these scores across the dataset to obtain a per-layer similarity profile for each condition.

3) *Hypotheses and Interpretation*: We test three ordered predictions.

a) *H1: Translation preserves the bridge better than paraphrasing within the same language*:

$$\text{sim}_{\text{strict}}^l > \text{sim}_{\text{para-eng}}^l. \quad (7)$$

This is the critical comparison because if a strict translation into a different language remains closer to the original representation than an English paraphrase, this suggests that the representation tracks content fidelity. In particular, it provides evidence that the information surviving translation is associated with the specific memorized fact.

b) *H2: Wording loss compounds with language change*:

$$\text{sim}_{\text{para-eng}}^l > \text{sim}_{\text{para-target}}^l. \quad (8)$$

Comparing the two paraphrase conditions isolates the additional effect of changing language once the wording has already been altered. A difference between these conditions indicates that language and wording fidelity both contribute to representational drift.

c) *H3: Strict translation remains closer than free-form paraphrasing in the target language*:

$$\text{sim}_{\text{strict}}^l > \text{sim}_{\text{para-target}}^l. \quad (9)$$

As a consequence of H1 and H2, we expect a faithful translation to produce a representation closer to the memorized English representation than a free-form paraphrase. This would indicate that content fidelity plays an important role in recalling the memorized segment.

We expect these differences to be more pronounced in the central layers, where representations may encode more abstract factual and associative content, and relatively minor in the initial layers, which are more closely associated with lexical and tokenization properties at the surface level and in the final layers, where representations have been observed to specialize in generating the target language [26].

A consistent ordering,

$$\text{sim}_{\text{strict}}^l > \text{sim}_{\text{para-eng}}^l > \text{sim}_{\text{para-target}}^l, \quad (10)$$

TABLE I
TRANSLATED SAMPLES FROM PILE-CC (200 PER PII TYPE) ARE NOVEL. WE REPORT DESCRIPTIVE STATISTICS OF THE WEB-PRESENCE SEARCH RESULTS FOR TWITTER HANDLES, EMAIL ADDRESSES AND PHONE NUMBERS. THE **QUERIES W/ RESULTS %** REPORTS THE PROPORTION OF QUERIES RETURNING AT LEAST ONE RESULT, WHILE **HISTORICAL CAPTURES PER QUERY** REPORTS THE NUMBER OF HISTORICAL CAPTURES IDENTIFIED FOR EACH QUERY. **EXACT MATCH** AND **MAX. FUZZY SCORE** DENOTES THE EXACT LEXICAL OVERLAP AND THE MAXIMUM FUZZY MATCHING SCORE AS DETAILED IN SECTION III-B1.

PII Type	Statistic	Queries w/ results %	Historical captures per query	Exact match	Max. Fuzzy score
Twitter	Mean	97.6	0.053	0	0.229
	Std.	15.2	0.615	0	3.237
	Min.	0	0	0	0
	25th percentile	100	0	0	0
	Median	100	0	0	0
	75th percentile	100	0	0	0
	Max.	100	13	0	46.289
Email	Mean	66.8	0.044	0	0.172
	Std.	47.1	0.693	0	2.800
	Min.	0	0	0	0
	25th percentile	0	0	0	0
	Median	100	0	0	0
	75th percentile	100	0	0	0
	Max.	100	14	0	47.327
Phone	Mean	100	0.054	0	0.163
	Std.	0	0.772	0	2.655
	Min.	100	0	0	0
	25th percentile	100	0	0	0
	Median	100	0	0	0
	75th percentile	100	0	0	0
	Max.	100	14	0	44.156

would be consistent with the existence of a **cross-lingual bridge: a representation anchored to specific memorized content** that remains **relatively stable under translation** but degrades when the content is substantially restated, particularly when paraphrasing is combined with a language change. This would explain why our TDE attacks transfer across languages while remaining sensitive to the fidelity of the attack prompt, and argues against a purely coarse-semantic account of the leakage, in which any topically related prompt, translated or not, would be equally effective.

IV. EXPERIMENTAL RESULTS

A. Web-Presence Check: Translated samples not Online

We check, on a subset of 200 samples per PII type, whether the translations used in our attacks were already available online before we generated them. If this were the case, a model might recognize our translation simply because it had already seen that exact text during training. Table I summarizes the results of our web verification pipeline (see Section III-B1).

a) Almost all queries returned some results, but almost none of these were actual matches: For all tested PII type, almost every query returned at least one page from the search engine (97.6% of queries for Twitter, 66.8% for emails, 100% for phone numbers). This simply means that the search terms were common enough to return *something*, but it does not mean that they are a match. When we examine how many of those returned pages actually contain a matching historical

screenshot, the numbers drop to almost zero: on average, only 0.053 matching captures per query for Twitter, 0.044 for emails and 0.054 for phone numbers. For the majority of the extracts there were no matching captures at all: in fact, the 25th percentile, the median and the 75th percentile are all 0. A small number of samples represented outliers, with a maximum of 13 (Twitter) or 14 (emails and phone numbers) matching hits, which brings the average slightly above zero, but the typical sample has none.

b) No online sample matches exactly: Out of the 200 samples checked for each type of PII, the exact match rate is 0 in every statistic we have reported: mean, minimum, maximum and every percentile.

c) Fuzzy matches are rare and, when they did occur, were in any case weak: The fuzzy match score measures how closely the text of a returned page matches our translated segment, taking into account minor differences in wording. For all the tested PII types, this score is 0 for the vast majority of segments (again, the 25th percentile, the median and the 75th percentile are all 0), which means that most of the returned pages are not significantly similar to our segments at all. The average score is low (0.229 for Twitter, 0.172 for emails, and 0.163 for phone numbers). Even the few outliers with higher scores (up to 46.3 for Twitter, 47.3 for email, 44.16 for phones) remain well below the similarity threshold we use to define something as a match (90, as described in Section III-B1).

These results demonstrate that the translated samples we checked are, for all intents and purposes, not present on the web: searches return numerous irrelevant pages, but virtually no exact matches, almost no historical copies, and only a handful of approximate matches that do not even come close to our match threshold. This supports our hypothesis that the translated data were not part of the models’ pre-training data and that the cross-linguistic leak we observe in Section IV-B reflects a genuine transfer of memorized content between languages.

B. Cross-Lingual Transferability of TDE Attacks

Having verified, on a sample of 200 passages per PII type, that the translations are not available online, we now check whether they can nevertheless cause a leak of private information across different languages. Table II reports, for each PII type, model, and language, the number of successfully leaked PII (Leak), the percentage of English leaks also observed in other languages (Rel.), and the total number of completions that contain a PII under the TDE attack.

TDE attacks transfer across different languages, and our results reveal that both the PII type and the multilingual capabilities of the model influence the attack success rate.

a) Leakage transfers to every target language, for every model: Across all seven models and three PII types, cross-lingual attacks in French, Italian, Spanish, and German recover a non-trivial number of PII instances that were only ever present, in their original form, in the English training data.

This confirms the central hypothesis of our study: memorization is not confined to the language in which the memo-

TABLE II

A LARGE PORTION OF PII ARE LEAKED ALSO IN OTHER LANGUAGES. WE REPORT THE NUMBER OF SUCCESSFULLY LEAKED PII, TOGETHER WITH THE PERCENTAGE OF ENGLISH LEAKS ALSO OBSERVED IN EACH LANGUAGE. THE NUMBER OF GENERATIONS CONTAINING PII INSTANCES IS REPORTED FOR EACH LANGUAGE.

PII type	Model	English	French			Italian			Spanish			German		
		Leak	Leak	Rel.	Generated	Leak	Rel.	Generated	Leak	Rel.	Generated	Leak	Rel.	Generated
Twitter	Llama 3.2 1B	340	255	75.0%	1788	243	71.5%	1724	260	76.5%	1893	213	62.6%	1369
	Llama 3.2 3B	548	492	89.8%	2284	501	91.4%	2327	488	89.1%	2334	389	71.0%	1831
	Qwen2.5 3B	468	395	84.4%	2330	372	79.5%	2360	384	82.1%	2391	298	63.7%	1693
	Qwen2.5 7B	591	479	81.0%	2682	439	74.3%	2647	440	74.5%	2626	361	61.1%	2197
	GPT-J 6B	1020	426	41.8%	2033	450	44.1%	2300	444	43.5%	2219	337	33.0%	1688
	GPT-Neo 1.3B	505	228	45.1%	1512	215	42.6%	1608	222	44.0%	1574	148	29.3%	1072
	GPT-Neo 2.7B	694	236	34.0%	1646	264	38.0%	1857	249	35.9%	1784	160	23.1%	1163
Email	Llama 3.2 1B	167	78	46.7%	1846	74	44.3%	2020	71	42.5%	1787	80	47.9%	2089
	Llama 3.2 3B	329	195	59.3%	2370	192	58.4%	2450	187	56.8%	2143	183	55.6%	2376
	Qwen2.5 3B	242	136	56.2%	2960	145	59.9%	2995	140	57.9%	2859	118	48.8%	2681
	Qwen2.5 7B	328	210	64.0%	3326	204	62.2%	3298	227	69.2%	3114	193	58.8%	3158
	GPT-J 6B	726	121	16.7%	2811	114	15.7%	2635	108	14.9%	2081	109	15.0%	2646
	GPT-Neo 1.3B	267	42	15.7%	1565	44	16.5%	1570	35	13.1%	1329	34	12.7%	1353
	GPT-Neo 2.7B	442	70	15.8%	2284	62	14.0%	2279	54	12.2%	1659	56	12.7%	2075
Phone	Llama 3.2 1B	161	32	19.9%	1851	37	23.0%	1691	35	21.7%	1793	6	3.7%	602
	Llama 3.2 3B	388	111	28.6%	2108	103	26.5%	1971	111	28.6%	2024	55	14.2%	1364
	Qwen2.5 3B	197	35	17.8%	2409	38	19.3%	2188	43	21.8%	2354	8	4.1%	235
	Qwen2.5 7B	285	54	18.9%	2443	58	20.4%	2340	61	21.4%	2450	10	3.5%	468
	GPT-J 6B	495	18	3.6%	1547	16	3.2%	1326	21	4.2%	1609	5	1.0%	233
	GPT-Neo 1.3B	138	3	2.2%	461	3	2.2%	472	3	2.2%	631	1	0.7%	91
	GPT-Neo 2.7B	256	4	1.6%	938	4	1.6%	837	6	2.3%	1021	1	0.4%	100

alized content was originally expressed, and the linguistic barrier offers only partial protection rather than a hard boundary.

b) Leakage is strongest in English but remains substantial cross-lingually, and this gap tracks a model’s multilingual capability: For every model, the English attack yields the highest leak count and leak rate, which is expected: it is the only condition where the attack prefix is drawn verbatim from a plausible training distribution.

However, the *size* of the drop between English and the other four languages differs sharply by model family. For the multilingual models (Llama 3.2, Qwen2.5), the cross-lingual leak rate retains up to 91.4% of the Twitter handles leaked in English, up to 69.2% for emails, and up to 28.6% for phone numbers. For the English-centric models (GPT-Neo, GPT-J), the same ratios are much smaller for Twitter handles (up to 45%) and email addresses (up to 16.7%), and collapse for phone numbers – most strikingly for GPT-J 6B and GPT-Neo 2.7B, which leak less than 3% of phone numbers cross-lingually.

This is consistent with our hypothesis that **cross-lingual leakage is mediated by shared multilingual representations**: models trained on substantially multilingual corpora exhibit a much smaller English-to-other-language gap, even though their absolute English leak rates are often lower than English-centric models’. Model scale alone does not explain this pattern – the largest English-centric model, GPT-J 6B, shows a *weak* transfer with respect to the similar-sized Qwen2.5 7B – which argues against transfer being a simple byproduct of overall model capability and in favor of it depending specifically on multilingual training.

c) PII type modulates both absolute leakage and transferability: Across all models, Twitter handles are leaked at far higher rates than emails, which in turn leak more than phone numbers, both in English and cross-lingually. This ordering likely reflects differences in the redundancy and predictability of each PII type’s surrounding context (handles are frequently repeated verbatim near associated usernames or mentions, while phone numbers are comparatively sparse and less syntactically constrained), consistent with prior evidence that extraction risk depends on structural properties of the target string as well as the language it is expressed in. Phone numbers are also the PII type for which cross-lingual transfer is proportionally weakest for the English-centric models.

These results establish that **TDE attacks transfer across the language boundary** for both monolingual and multilingual models, that the **degree of transfer scales with a model’s multilingual training** and that this cross-lingual susceptibility is not uniform but is shaped by the interaction of model family, language, and PII type.

C. Transferability of TDE Attacks Under Paraphrase

Section IV-B showed that TDE attacks transfer across languages when the attack prefix is a faithful, strict translation of the memorized English passages. We therefore ask what factors are key to the effective retrieval of memorized PII. Building on the disentanglement of language and wording fidelity introduced in Section III-D, we now first test its behavioral counterpart: whether leakage itself, not just representational similarity, tracks content fidelity.

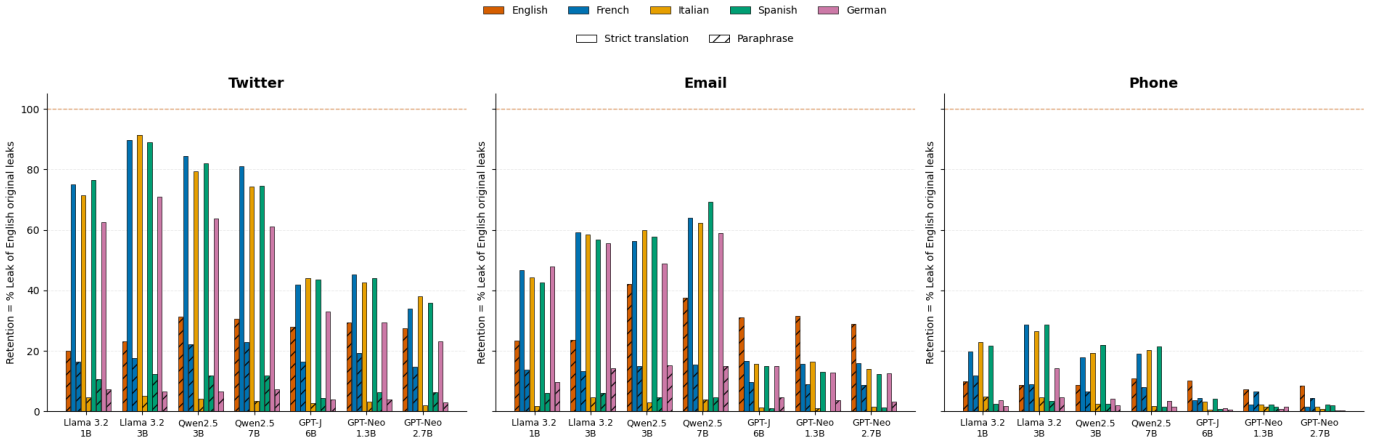


Fig. 1. Retention of leaked PII instances under a paraphrased attack, compared to strict-attack leak count, averaged across the four multilingual models (Llama 3.2 1B/3B, Qwen2.5 3B/7B) and the three English-centric models (GPT-J 6B, GPT-Neo 1.3B/2.7B)

We hence first analyze the behavioral effects of paraphrases, performing the TDE attacks on the paraphrased inputs. Figure 1 reports, for the same PII types and models as Table II, the leakage obtained when the attack prefix is paraphrased. To facilitate comparisons across languages and models, we report leakage as a percentage of the corresponding English leaks for each model and PII type. Thus, 100% indicates the same number of leaked PII instances as the English baseline, while values below 100% indicate lower leakage. We refer to this relative leakage as *retention* (for completeness, in Appendix B we report the raw number of leaks).

a) *Paraphrasing the attack sharply reduces leakage, in every language and for every model:* For both Twitter handles and email addresses, the retention rate for paraphrased English input drops to about one-quarter or one-third of the count obtained with the strict attack, while the retention rate for paraphrased input in the target language drops even further, to about one-tenth. For phone numbers, the results vary more widely, but the number of leaks when using paraphrased input is always lower than the corresponding number of leaks obtained with strict translation for multilingual models, and comparable in English-centric models (which leak only a small fraction of the phone numbers). Furthermore, for all multilingual models, the leak rate in the case of a paraphrasing attack is substantially lower than that obtained with literal translation into other languages. These results support our hypothesis that the loss of the exact wording leads to a substantial loss of extraction success, even when the target language does not change. We will explore this point further in Section IV-D.

b) *English paraphrase consistently retains more leaked instances than cross-lingual paraphrase:* English paraphrase retention is roughly two to three times higher than cross-lingual paraphrase retention, for both multilingual and English-centric models alike, with the only exception registered on phone numbers extracted on Llama 3.2 1B. This is the attack-level counterpart of the Strict > English paraphrase > Target

language paraphrase ordering that we expect to hold representationally as well, as described in Section III-D: paraphrasing costs leakage, and paraphrasing into *another language* costs more of it than paraphrasing alone.

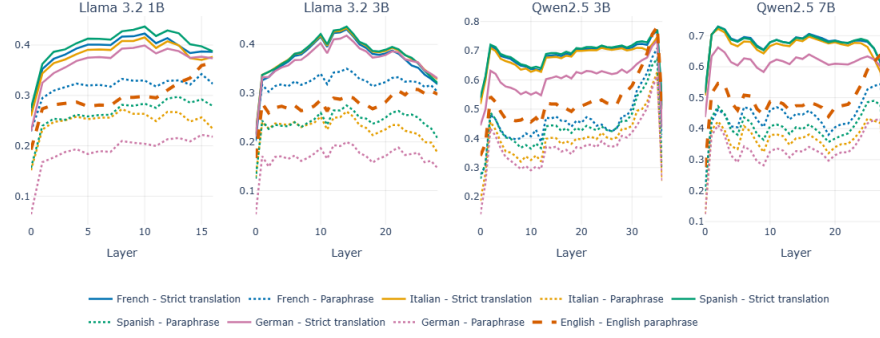
Paraphrasing the attacker’s prompt substantially reduces training-data extraction, and reduces it further when combined with translation into another language – consistent with the content-fidelity ordering hypothesized in Section III-D. We hence expect representational similarity to track the same content-fidelity ordering, which we test directly in the next section.

D. Representational Evidence for Cross-Lingual Bridges

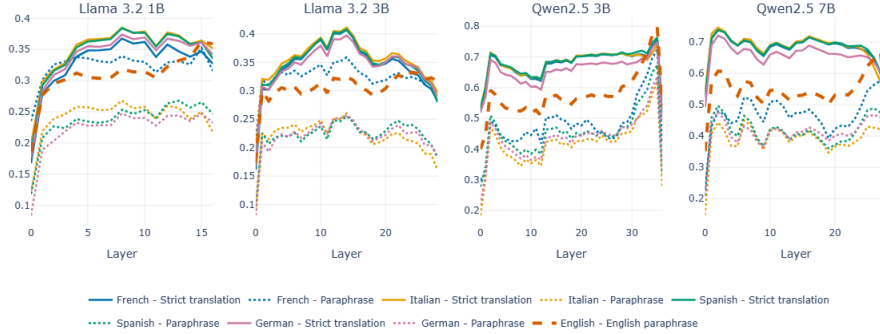
To explain the success of TDE attacks in other languages, we then demonstrate that, from a representational point of view, models tend to *bridge* structurally similar inputs from different languages. As described in Section III-D, we hence compare the similarity of the representations under strict translations and free-form paraphrases. For this analysis, we focus on the activations of multilingual LLMs, since TDE attacks are more likely to transfer for these models.

Figure 2 report the per-layer results of the similarity analysis described in Section III-D, for Twitter handles (2-a), email addresses (2-b) and phone numbers (2-c) respectively. For each model, we plot the mean cosine similarity to the English original representation across layers, for the strict translation for each target language, the English paraphrase, and the target-language paraphrase. Table III summarizes the same results averaged across all layers, for each model, PII type, and condition, reporting the German strict translation separately from the French, Italian and Spanish ones.

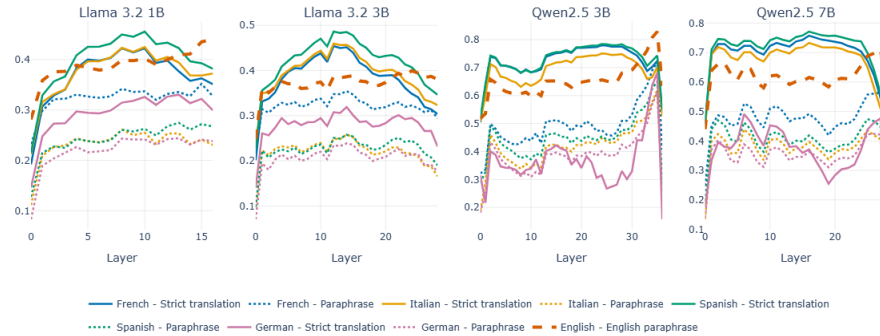
a) *The predicted ordering holds for Twitter handles and emails, and for phone numbers except in German:* As we discussed in Section III-D, if the models bridge different languages’ representations, three conditions should hold: strict translation should preserve greater similarity to the English original than an English paraphrase (H1: Equation 7), an English paraphrase should preserve greater similarity than



(a) Twitter



(b) Email



(c) Phone

Fig. 2. Mean cosine similarity between the representations of each strict translation and paraphrase and the English original, by layer, for each PII type.

a target-language paraphrase (H2: Equation 8), and strict translation should consequently exceed the target-language paraphrase as well (H3: Equation 9). For Twitter handles and emails, all three hold for every one of the four models, for both the French/Italian/Spanish translations and the German one, when similarity is averaged over layers: strict translation > English paraphrase > target language paraphrase in every row of Table III. For phone numbers, H2 holds in every row, while H1 and H3 hold for French, Italian and Spanish (for H1, in three of the four models; Llama 3.2 1B is on par, 0.380 vs 0.389). German is the exception: its strict translation is below

the English paraphrase at every layer and in every model, and, for Qwen2.5, below the average target-language paraphrase as well (0.365 vs 0.434 and 0.369 vs 0.399). Where it holds, this ordering is consistent with **a bridge anchored to content fidelity**: representational similarity degrades least when only the language changes (strict translation), more when only the wording changes (English paraphrase), and most when both change simultaneously (target language paraphrase).

b) Multilingual models show a substantially stronger bridge than English-centric ones: The gap between conditions and the absolute similarity level scale with the model's

TABLE III

MEAN COSINE SIMILARITY TO THE ENGLISH REPRESENTATION, AVERAGED OVER ALL LAYERS. STRICT TRANSLATIONS ARE REPORTED SEPARATELY FOR FRENCH/ITALIAN/SPANISH (MEAN OVER THE THREE LANGUAGES) AND GERMAN; PARAPHRASES ARE AVERAGED OVER THE FOUR TARGET LANGUAGES.

PII type	Model	Strict (FR/IT/ES)	Strict (DE)	English paraphrase	Target language paraphrase
Twitter	Llama 3.2 1B	0.389	0.366	0.296	0.255
	Llama 3.2 3B	0.375	0.367	0.279	0.239
	Qwen2.5 3B	0.681	0.603	0.524	0.417
	Qwen2.5 7B	0.668	0.607	0.491	0.381
Email	Llama 3.2 1B	0.340	0.338	0.309	0.255
	Llama 3.2 3B	0.348	0.343	0.305	0.246
	Qwen2.5 3B	0.678	0.653	0.575	0.447
	Qwen2.5 7B	0.679	0.652	0.546	0.416
Phone	Llama 3.2 1B	0.380	0.293	0.389	0.254
	Llama 3.2 3B	0.398	0.279	0.372	0.243
	Qwen2.5 3B	0.714	0.365	0.652	0.434
	Qwen2.5 7B	0.695	0.369	0.619	0.399

multilingual training. The two Llama models – despite being described as multilingual – stay below 0.4 mean cosine similarity in every condition, and the layer-averaged difference between strict translation (French/Italian/Spanish) and target-language paraphrase is comparatively narrow (0.09–0.16). Qwen2.5, trained on a substantially more multilingual corpus, reaches mean similarities of 0.67–0.71 for the strict condition and shows a much wider separation between conditions (a Strict/target language paraphrase gap of roughly 0.23–0.30). This mirrors the extraction-transferability results in Section IV-B: the same models whose leak rates transfer most completely across languages are also the ones whose internal representations show the clearest content-anchored bridge.

c) The bridge is present for all three PII types, but its strength does not follow the leakage ordering: The Strict/target language paraphrase gap (French/Italian/Spanish strict translations) is 0.13–0.29 for Twitter handles, 0.09–0.26 for emails and 0.13–0.30 for phone numbers (Table III). The gap is smallest for emails in every model, and comparable between Twitter handles and phone numbers. Unlike the leakage rates presented in Section IV-B, where Twitter handles are leaked most and phone numbers least, the strength of the bridge therefore does not order the PII types consistently, and we do not read it as a proxy for extraction risk across PII types.

Twitter handles also show a distinct pattern: target-language paraphrases sit closer to each other (0.04–0.11 apart) than either does to strict translation (0.09–0.29 apart).

German phone numbers are the exception to the ordering above: their strict translation reaches only 0.28–0.37 mean similarity (versus 0.38–0.71 for French, Italian and Spanish), it lies below the English paraphrase at every layer, and it is the only strict condition that falls below the target-language paraphrase in some models. This coincides with the sharp drop in the number of generations containing a PII and of leaks for German phone numbers (Table II), suggesting that this condition behaves differently from the others; we therefore report it separately in Table III.

d) Representations are bridged in the middle layers, and then diverge: Across all four models, the layer-wise curves trace a consistent arc: similarity to the English original is lowest at the input end (and, for Qwen2.5, also at the output end), and highest in between, where the cross-lingual bridge appears to be located. At the embedding layer (layer 0), before any contextual processing has occurred, similarity is driven mostly by shared surface tokens. For Twitter handles and emails, the predicted ordering (strict > eng-paraphrase > L2-paraphrase) reasserts itself within the first few transformer layers and holds through the middle of the network (in at least 97% of the middle layers), which we take as the region where content-level, language-agnostic representations are formed. For phone numbers, this ordering holds less consistently (75–96% of the middle layers for French, Italian, and Spanish), while the German strict translation stays below the English paraphrase at every layer.

Toward the output end, the bridge again gives way: for Qwen2.5 in particular, all three conditions – strict, eng-paraphrase, and L2-paraphrase alike – show a pronounced drop in absolute similarity over the final one to three layers (a fall of roughly 0.12–0.35 in cosine similarity from each curve’s peak). We read this late-layer divergence as a property of the output layers generally, not of the bridge specifically: the last layers of a language model are known to specialize toward predicting the next token in the correct surface language, which pulls every condition’s representation away from the English original by a similar amount. Critically, the target-language paraphrase remains the furthest from the English original at every layer, all the way to the final one, while the strict translation and the English paraphrase come close to each other (at the very last layer the English paraphrase is on par with or slightly above the strict translation for emails and phone numbers). The late-layer drop therefore rescales the overall level of similarity without erasing the separation between the target-language paraphrase and the other conditions, which is part of the evidence for the bridge. Taken together, the arc suggests that cross-lingual leakage is enabled by a mid-network representation that is comparatively insulated from both the surface form of the input and the language-specific demands of output generation.

The representational analysis corroborates the attack-level transferability results: a language-crossing bridge exists, it is anchored to content fidelity instead of topic alone (for Twitter handles, emails and French/Italian/Spanish phone numbers, translation beats paraphrase, even a same-language one), and its strength scales with a model’s multilingual training, not its parameter count. Its strength across PII types, however, does not mirror the leakage counts in Table II, and the German phone-number condition remains an open exception.

V. DISCUSSION AND CONCLUSIONS

This paper set out to test an assumption implicit in most prior work on training data extraction: that an attacker queries a model in the same language the memorized content was originally written in. Our results show this assumption does

not hold. TDE attacks transfer across the language boundary for both English-centric and multilingual models, a sample of the translated passages (200 per PII type) was verified to be absent from the web prior to our experiments, and the degree of transfer tracks a model’s multilingual training.

By independently varying language and wording fidelity, we further showed that this leakage is anchored to the specific memorized content: a strict translation recovers substantially more leakage than a paraphrase in either English or the target language, and, for Twitter handles and emails, also produces a more similar internal representation (for phone numbers, this holds for French, Italian and Spanish but not for German).

Representationally, this content-anchored bridge is neither present from the input embeddings nor preserved into the output layers, but is concentrated in a band of middle transformer layers, consistent with the view that these layers encode language-agnostic, fact-level associations before the final layers specialize toward generation in the surface language of the prompt [26].

a) Connection to cross-lingual unlearning: These results offer a plausible mechanistic account for a finding we could previously only cite, not explain: [18] show that unlearning a fact in one language does not remove it in another – a deletion in English typically leaves the same fact recoverable, verbatim, in Spanish or Chinese. If the cross-lingual bridge we identify is concentrated in the middle layers, while existing unlearning and model-editing methods largely target specific factual associations localized around particular parameters or layers, it is plausible that such methods remove the language-specific surface association without touching the shared mid-layer representation that the bridge depends on. Under this account, cross-lingual leakage after unlearning would not be a failure of unlearning *per se*, but a symptom of editing a representation at the wrong depth. We see directly testing this account – by applying unlearning or editing methods and then repeating our layer-wise similarity analysis – as a natural and high-value extension of this work.

b) Practical implications: For practitioners, the central implication is that monolingual privacy evaluation is not sufficient for a multilingual model. A model that appears to resist an English-language extraction attempt cannot be assumed to resist the same attempt translated into another language, and our results suggest the opposite is more likely as multilingual training becomes more extensive. This has direct bearing on red-teaming and compliance practice: privacy audits, safety filters, and memorization benchmarks that are evaluated only in a model’s dominant language will systematically underestimate real-world exposure for any model with meaningful multilingual capability – which, at this point, describes most production-grade LLMs.

c) Limitations: Our target languages – French, Italian, Spanish, and German – are all high-resource and, with the partial exception of German, typologically close to English. This was a deliberate choice to isolate the language-versus-wording effect under favorable translation quality, but it leaves open, and likely understates, the risk for lower-resource or

typologically distant languages, where prior work suggests both weaker safety alignment [5] and different memorization dynamics [16], [19], [20] may amplify instead of dampen the effect we observe. Second, our phone-number results rest on small absolute leak counts, particularly under paraphrase and for the English-centric models, and should be interpreted with corresponding caution. In particular, the German phone-number condition behaves differently from the other languages, both behaviorally (Table II) and representationally (Table III, Fig. 2), and we report it separately.

d) Future work: Beyond directly testing the unlearning connection above, three extensions follow naturally from our limitations. First, repeating this study on lower-resource and typologically distant target languages would test whether the cross-lingual bridge strengthens, weakens, or changes character outside the high-resource, Western-European setting studied here. Second, extending the paraphrase-based disentanglement methodology from extraction to jailbreaking would let us test whether the cross-lingual latent space that [5] exploit for safety circumvention has the same content-anchored, middle-layer structure we find for memorization, or whether the two phenomena rely on different mechanisms despite superficially similar symptoms. Third, our similarity-based evidence for the bridge is correlational at the representation level; combining it with causal interventions – for instance, patching or ablating the token-level contributions we identify as most responsible for cross-lingual similarity – would let us move from showing that a bridge exists to showing precisely which components of the model implement it.

In summary, we provide the first direct evidence that PII memorized from English training data can be recovered by attacks formulated in other languages, that this transfer is driven by a genuine, content-anchored cross-lingual bridge and that the strength of this bridge scales with a model’s multilingual training. As multilingual deployment continues to expand, we argue that privacy evaluation, auditing, and mitigation must be treated as inherently cross-lingual problems, not extended to other languages as an afterthought.

OPEN SCIENCE

We release the full code required to reproduce our experiments (the TDE attack pipeline, the translation and paraphrase generation scripts, the web-presence verification pipeline, and the analysis and plotting notebooks used to produce every table and figure in this paper) via a fully anonymized repository: <https://anonymous.4open.science/r/TransferTDE-CC84/>. This repository will remain accessible throughout the review process. We do not release the underlying dataset itself – the English passages, and their translations and paraphrases – as it consists of samples containing real, identifiable personal information (email addresses, phone numbers, and Twitter handles) drawn from *Pile-CC*, and redistributing it would work against the privacy considerations (discussed in our Ethical Considerations section). We will make the constructed dataset available to researchers upon request for legitimate research

purposes, subject to a data use agreement and institutional verification.

LLM USAGE CONSIDERATIONS

a) Accountability and Correctness: LLMs were used for editorial purposes in this manuscript, and all outputs were inspected by the authors to ensure accuracy and originality. The authors reviewed and edited all LLM-generated content used in the paper, including generated code, as if it had been written by the authors themselves, and take full responsibility for its accuracy and integrity. Beyond editorial use, an LLM (Gemma-3-27B-it, accessed via the Google Gemini API) was used methodologically, as a core component of our data pipeline, to generate the translations described in Section III-B. The paraphrases used in Section III-D were instead generated with open-weight models run locally (Appendix A).

At least one author proficient in each target language conducted spot checks of the translations and paraphrases, and we additionally verified the integrity of the paraphrases in Appendix A. No experimental results, references, or claims in this paper were generated by an LLM without independent verification against the underlying data or source material.

b) Transparency: The use of an LLM to generate translations is an integral part of our methodology, not a secondary element: our experimental design relies on the availability of faithful, structure-preserving translations of a dataset of the required scale, which we obtained by instructing Gemma-3-27B-it to that effect (Section III-B). The free-form paraphrases, which provide the contrasting fidelity condition (Section III-D), were generated with open-weight models run locally (Appendix A). We independently validated the results of this process. We acknowledge a limitation that this choice introduces: since Gemma-3-27B-it is accessed via a hosted API with versioned weights, our translation generation phase is not perfectly reproducible if the underlying model provided by the API undergoes changes or is deprecated in the future. The paraphrases are not affected by this limitation, as they were produced with locally-run open-weight models. We mitigate this issue by publishing, in our artifacts, the exact prompts and scripts used to generate the translations and paraphrases in this article (subject to the data-sharing constraints described in our Open Science section).

c) Responsibility: The target models evaluated in this study (GPT-Neo 1.3B/2.7B, GPT-J 6B, Llama 3.2 1B/3B, Qwen2.5 3B/7B) are the *subjects* of our study; we did not train any models, nor did we collect new data beyond the translation and paraphrasing of pre-existing texts from Pile-CC (Section III-B). We selected models covering two parameter ranges (approximately 1 to 3B and 6 to 7B) within each family specifically to verify whether cross-lingual transfer is proportional to model size or to the composition of the multilingual pre-training; our results point to the latter hypothesis, which justified our decision to limit our evaluation to this range and not to more computationally intensive models. All inference on the target models was performed using greedy decoding, limiting each query to a single forward generation step. The

dataset used by PII type and language was restricted to obtain statistically significant leakage counts without querying the models more than necessary for this purpose. The experiments were run on a single RTX A6000 from a private, local server. Each inference took from 10s/it for smaller models up to 20s/it for larger ones. Considering also the large number of hours needed to perfect the experimental setup, we estimate a total of 100 hours of GPU usage. The estimated emissions are 12.96 kg of CO₂.³

ETHICAL CONSIDERATIONS

This paper analyzes whether personally identifiable information (PII) stored by large language models during pre-training can be extracted using prompts formulated in a language other than the one in which the information was originally recorded. Below, we discuss the risks this entails and the measures we have taken to mitigate them.

a) Data provenance and handling: All PII used in our experiments comes from Pile-CC, a pre-existing dataset derived from Common Crawl. We did not collect, extract, or generate any new PII; the email addresses, phone numbers, and Twitter handles we analyze were already present in this corpus and, by extension, already present in the training data of the models we evaluate, prior to our intervention. Our translations and paraphrases are transformations of this existing data, not new instances of it. We did not attempt to identify, contact, or otherwise interact with the individuals associated with the PII in our dataset, and we report our results in aggregate form (number of data leaks and similarity statistics). The English extracts, which contain PII, were processed by two third-party services: (i) the Google Gemini API, to which we sent the extracts to obtain the translations (Section III-B), and (ii) the Tavily search API, to which we sent queries derived from the translated extracts for the web-presence check (Section III-B1). No PII was sent to third-party services for paraphrase generation, which was performed locally (Appendix A). We do not publish, list, or otherwise reproduce any of the individual instances of PII retrieved during our experiments. Such information might be shared for research purposes only by contacting directly the corresponding authors.

b) Dual-use considerations: Attacks based on the extraction of training data are, by their very nature, dual-use: the same techniques that allow us to measure a privacy risk could, in principle, be used by a malicious actor to materialize that risk. We believe that disclosing the results we have obtained promotes the development of new defense techniques for two reasons. First, the underlying vulnerability (monolingual memorization and verbatim retrieval) has already been well established in previous work; our contribution lies in demonstrating that current monolingual extraction and evaluation practices are insufficient, which in itself constitutes an argument in favor of more thorough defense and oversight. Second, concealing this finding would prevent model providers

³Estimate computed on <https://mlco2.github.io/impact/>

and downstream implementers from realizing that monolingual privacy evaluation, red-teaming, and mitigation measures (e.g., unlearning or differential privacy applied and tested in only one language) may substantially underestimate real-world exposure, which, in our view, represents a greater harm than the marginal advantage our results provide to an attacker who could, in any case, already translate a known extraction prompt on their own.

c) Responsible disclosure: We did not pursue formal disclosure, as all the models evaluated are open-weight, and our findings pertain to a general property of multilingual training. We encourage practitioners who apply unlearning techniques, differential privacy, or other mitigation measures to evaluate their effectiveness across languages.

d) Scope of risk: The types of PII we study (email addresses, phone numbers, and Twitter handles) are all categories of information that may already be publicly available on the web in the specific cases we extracted from Pile-CC, and our web presence check (Section III-B1) was designed to verify the novelty of the translation for the purposes of our methodology, not to assess whether the underlying English-language PII remains publicly identifiable through other means. We did not evaluate the extraction of more sensitive categories of PII (e.g., government-issued identifiers, financial or health information), and we caution against extrapolating our leakage rate estimates to such categories without a separate evaluation.

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A. Paraphrase Generation and Evaluation

To construct the variants x_{para}^L used in the cross-lingual bridge analysis of Section III-D, we generate, for each sample and each target language, a paraphrased version.

1) *Paraphrase Generation*: For each language, we use an open-weight instruction-tuned language model, selected to maximize paraphrase quality in that specific language. Table IV lists the model used for each language. For German, we made two attempts to improve paraphrase quality, using first a language-specific model and then a general-purpose one (see the German model comparison below); Table IV reports the model retained. All models are run locally, so no PII-containing sample is transmitted to a third-party API during paraphrase generation.

TABLE IV
MODELS USED FOR PARAPHRASE GENERATION.

Language	Model
English	Qwen2.5-7B-Instruct
German	Qwen2.5-7B-Instruct
French	Mistral-7B-Instruct-v0.3
Italian	Minerva-7B-instruct-v1.0
Spanish	Salamandra-7B-instruct

Each model is prompted with an instruction that requests a copy of the input sample in the same language, explicitly requiring that all entities relevant to the attack (email addresses, phone numbers, handles, proper names, dates, and URLs) be preserved word-for-word. This constraint is necessary because the paraphrased samples are subsequently used as alternative attack prompts, so any unintended alteration of the target PII would invalidate the downstream extraction experiment.

Generation uses sampling with temperature 0.7 and top- p 0.9, with a maximum of 512 new tokens per completion.

2) *Evaluation Metrics*: We evaluate the generated paraphrases along two complementary axes: lexical divergence from the original sample and preservation of semantic content. A good paraphrase should score *low* on the former and *high* on the latter.

Lexical divergence. We compute sentence-level BLEU [27] (via `sacrebleu`) and ROUGE-L F1 [28] (via the `rouge-score` implementation) between each original sample and its paraphrase. Low values in both metrics indicate that the paraphrase does not merely reproduce the source text.

Semantic similarity Lexical divergence alone does not guaranty that meaning is preserved, since a phrase could diverge too far from the original content. We therefore additionally measure semantic equivalence using a multilingual Natural Language Inference (NLI) model, `xlm-roberta-large-xnli` [29], fine-tuned on XNLI and covering all five languages used in this study. For each pair

(context, context) we compute the entailment probability in both directions and average them:

$$\text{sim} = \frac{1}{2} \left(P_{\text{entail}}(\text{context} \Rightarrow \text{paraphrase}) + P_{\text{entail}}(\text{paraphrase} \Rightarrow \text{context}) \right). \quad (11)$$

A high bidirectional entailment score indicates that the two texts imply one another and are therefore likely to be semantically equivalent, whereas a one-directional entailment could instead reflect a generalization or specialization of the original content.

3) *Results*: Table V reports, for each language of PII, the mean BLEU, ROUGE-L, and semantic similarity between the original samples and their paraphrases.

a) *German Paraphrase Model Comparison*: For German, we made two attempts to improve the quality of the paraphrases. In a first attempt, we used DiscoLM German 7b v1, a Mistral-based model fine-tuned specifically for German. In a second attempt, we used Qwen2.5-7B-Instruct, the general-purpose model employed for English, on the same German samples. Table VI reports the comparison. The second attempt yields substantially higher bidirectional semantic similarity (about 90% versus 61–70% across PII types), at the cost of a somewhat higher lexical overlap with the original (higher ROUGE-L for all PII types, and higher BLEU for emails). Since preserving the meaning of the original extracts is essential for the paraphrase to be a valid attack prompt, we retained the paraphrases generated with Qwen2.5-7B-Instruct for all German results reported in Section III-D and in the experimental results.

B. Raw Results for Paraphrase Attacks

Table VII reports the absolute number of leaked PII instances obtained from the TDE attacks across models, languages, PII types, and input conditions. For each language, we report the leakage obtained with a strict translation of the original attack prefix and with its paraphrased counterpart. These raw counts provide the basis for the relative leakage analysis presented in Figure 1, where leakage is normalised to the corresponding English original prefix for each model and PII type.

TABLE V
MEAN BLEU, ROUGE-L (F1), AND BIDIRECTIONAL SEMANTIC SIMILARITY BETWEEN ORIGINAL EXCERPTS AND THEIR PARAPHRASES, BY PII TYPE AND LANGUAGE. LOWER BLEU/ROUGE-L AND HIGHER SEMANTIC SIMILARITY INDICATE BETTER PARAPHRASE QUALITY.

PII type	Language	Model	BLEU	ROUGE-L	Sem. sim.
Email	English	Qwen2.5-7B-Instruct	23.51	49.21	83.43
	French	Mistral-7B-Instruct-v0.3	8.21	15.02	69.23
	Italian	Minerva-7B-instruct	22.08	39.94	46.99
	Spanish	Salamandra-7B-instruct	6.62	18.84	41.35
	German	DiscoLM_German_7b	41.89	56.41	68.05
	German	Qwen2.5-7B-Instruct	47.80	67.70	90.89
Phone	English	Qwen2.5-7B-Instruct	27.05	52.87	84.18
	French	Mistral-7B-Instruct-v0.3	7.34	14.88	66.65
	Italian	Minerva-7B-instruct	22.58	40.98	48.88
	Spanish	Salamandra-7B-instruct	6.44	20.16	42.10
	German	DiscoLM_German_7b	49.31	62.21	69.85
	German	Qwen2.5-7B-Instruct	41.98	63.94	90.70
Twitter	English	Qwen2.5-7B-Instruct	18.96	44.85	83.56
	French	Mistral-7B-Instruct-v0.3	6.06	13.70	75.38
	Italian	Minerva-7B-instruct	18.33	36.28	47.06
	Spanish	Salamandra-7B-instruct	7.48	22.93	54.36
	German	DiscoLM_German_7b	45.20	56.36	61.46
	German	Qwen2.5-7B-Instruct	43.12	64.79	92.02

TABLE VI
COMPARISON OF TWO CANDIDATE PARAPHRASE MODELS FOR GERMAN.

PII type	Model	BLEU	ROUGE-L	Sem. sim.
Email	DiscoLM_German_7b	41.89	56.41	68.05
	Qwen2.5-7B-Instruct	47.80	67.70	90.89
Phone	DiscoLM_German_7b	49.31	62.21	69.85
	Qwen2.5-7B-Instruct	41.98	63.94	90.70
Twitter	DiscoLM_German_7b	45.20	56.36	61.46
	Qwen2.5-7B-Instruct	43.12	64.79	92.02

TABLE VII
NUMBER OF LEAKED PII INSTANCES FOR THE STRICT TRANSLATION AND PARAPHRASED ATTACK PREFIXES. RESULTS ARE REPORTED FOR EACH MODEL, LANGUAGE, AND PII TYPE.

Model	English		French		Italian		Spanish		German	
	Strict	Paraphrase	Strict	Paraphrase	Strict	Paraphrase	Strict	Paraphrase	Strict	Paraphrase
Twitter										
Llama 3.2 1B	340	68	255	56	243	16	260	36	213	25
Llama 3.2 3B	548	127	492	96	501	28	488	68	389	36
Qwen2.5 3B	468	146	395	104	372	19	384	55	298	31
Qwen2.5 7B	591	181	479	135	439	20	440	70	361	43
GPT-J 6B	1020	284	426	168	450	28	444	44	337	40
GPT-Neo 1.3B	505	148	228	97	215	16	222	32	148	20
GPT-Neo 2.7B	694	190	236	103	264	14	249	44	160	21
Email										
Llama 3.2 1B	167	39	78	23	74	3	71	10	80	16
Llama 3.2 3B	329	78	195	44	192	15	187	20	183	47
Qwen2.5 3B	242	102	136	36	145	7	140	11	118	37
Qwen2.5 7B	328	123	210	51	204	13	227	15	193	49
GPT-J 6B	726	226	121	71	114	9	108	7	109	33
GPT-Neo 1.3B	267	84	42	24	44	3	35	0	34	10
GPT-Neo 2.7B	442	128	70	39	62	7	54	6	56	14
Phone										
Llama 3.2 1B	161	16	32	19	37	8	35	4	6	3
Llama 3.2 3B	388	34	111	35	103	18	111	13	55	18
Qwen2.5 3B	197	17	35	13	38	5	43	5	8	4
Qwen2.5 7B	285	31	54	23	58	5	61	4	10	4
GPT-J 6B	495	50	18	22	16	3	21	4	5	3
GPT-Neo 1.3B	138	10	3	9	3	2	3	2	1	2
GPT-Neo 2.7B	256	22	4	11	4	2	6	5	1	1